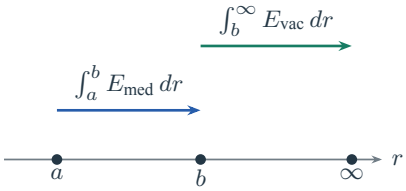
M3-006b 由分段径向场求导体球电势



介质段 $a < r < b$　真空段 $r > b$

积分从导体表面沿径向走向无穷远；$\varphi(\infty) = 0$

$$\varphi(a) = \int_a^b E_{\text{med}}dr + \int_b^\infty E_{\text{vac}}dr$$

$$E_{\text{med}} = \frac{Q}{4\pi\varepsilon_0\varepsilon_r r^2}$$

$$E_{\text{vac}} = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$C = Q/\varphi(a)$$